Supercritical Inverse-Square Dynamics: Complete Spectral Resolution and Scattering

HCS-C391 theorem and reproducibility package

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Abstract

We give a convention-complete spectral reconstruction of every unit-modulus boundary realization of the supercritical half-line inverse-square Hamiltonian. The classical collision flow and its quantum self-adjoint extension remain distinct sources of time evolution. An explicit Green function yields the entire bilateral negative ladder, while its positive boundary jump gives a continuous spectrum of multiplicity one and excludes hidden singular spectrum. Incoming waves are normalized to $(2\pi)^{-1/2}\exp(-ikx)$. The resulting reflection amplitude has unit modulus, but differs by a minus sign from scattering relative to the free Dirichlet half-line. The hypotheses of the classical complete-wave-operator theorem are checked for every positive coupling parameter and every unit boundary phase. Independent high-precision tests compare 108 Green-kernel jumps with the spectral-density formula. These tests audit constants and phases; completeness comes from endpoint and spectral analysis. No target arithmetic interpretation or novel ownership of the classical inverse-square spectrum is claimed.

Keywords: inverse-square Hamiltonian; self-adjoint domain; bilateral bound spectrum; continuous scattering; discrete scaling; determinant obstruction.

中文摘要：
本文在全部自伴边界与负能级结论之外，加入完整连续谱分解和散射规范。
格林核的边界跳跃确定单重绝对连续谱；入射波归一化明确区分反射振幅与相对散射。
独立高精度核验直接比较谱密度与格林核跳跃，防止常数和相位错误。
完备性来自算子证明及已核实条件的文献定理，而非有限计算。
关键词：反平方势；自伴定义域；双向负谱；连续散射；离散尺度；行列式障碍

1 A local Hamiltonian does not choose a boundary

Fix $\sigma > 0$, put $g = \sigma^2 + 1/4$, and use units $\hbar = 2m = 1$. The classical source is $h(x, p) = p^2 - g/x^2$ on $T^*(0, \infty)$. Its quantum expression is $\ell = -\partial_x^2 - g/x^2$ on $\mathcal{H} = L^2((0, \infty), \mathrm{d}x)$. A unit complex number κ will specify the missing boundary condition. It is independent physical input, not a value fitted to a desired eigenvalue or prime scale. Throughout, I_ν, J_ν, K_ν are ordinary Bessel functions.

The contribution is a single-convention source theorem and audit, not a new discovery of the inverse-square spectrum. Dereziński and Richard give the domain, spectrum and scattering theory in a substantially larger complex parameter family [1]. We retain the entire strictly supercritical self-adjoint subfamily, derive the constants used here, and connect them to the original source clock and the limits of ordinary global determinants. The critical value $\sigma = 0$, nonunit κ, ultraviolet regularization and confinement are not part of this theorem.

Proposition 1 (The complete collision clock)**.** *For every initial point (x_0, p_0), with energy $E = h(x_0, p_0)$,*

$$y(t) = x(t)^2 = x_0^2 + 4x_0p_0t + 4Et^2, \qquad p(t) = \frac{y'(t)}{4\sqrt{y(t)}}. \tag{1}$$

Its maximal time interval is the component of $\{t : y(t) > 0\}$ containing zero. Every orbit has a finite collision endpoint, and none is periodic in the declared phase space.

Proof. Hamilton's equations are $x' = 2p$ and $p' = -2g/x^3$, so $y'' = 8E$. The reconstruction in (1) satisfies both equations. For $E \neq 0$ the quadratic discriminant is $16g > 0$. At $E < 0$ its positive component is bounded; at $E > 0$ it is a half-line. At $E = 0$ the nonzero slope is $\pm 4\sqrt{g}$, again giving a half-line. Each finite endpoint has $x = 0$, which is excluded. No collision continuation has been added, so none of these intervals gives a periodic orbit. □

2 All self-adjoint domains and the bound ladder

Let $D_{\max} = \{f \in L^2 : \ell f \in L^2$ distributionally$\}$ and let $D_{\min}$ be the graph closure of $C_c^\infty(0,\infty)$. Membership in $D_{\min}$ near zero means membership after multiplication by a smooth cutoff equal to one there. Define

$$D_\kappa = \{f \in D_{\max} : f - c(\kappa x^{1/2-\mathrm{i}\sigma} + x^{1/2+\mathrm{i}\sigma}) \in D_{\min} \text{ near } 0 \text{ for some } c \in \mathbb{C}\}. \tag{2}$$

Theorem 1 (Exhaustive boundary family)**.** *The domains* (2)*,* $|\kappa| = 1$*, are exactly the self-adjoint extensions of the minimal operator. None is semibounded, and there is no Friedrichs realization. Position-space complex conjugation preserves every one of these domains.*

Proof. The zero-energy basis $u_\pm = x^{1/2\pm\mathrm{i}\sigma}$ is square integrable near zero. Variation of constants gives a unique expansion $f = a_f u_- + b_f u_+$ modulo $D_{\min}$ there. Indeed, the integrals of $u_\pm \ell f$ converge by Cauchy–Schwarz and their remainders are $o(x^{3/2})$, with derivatives $o(x^{1/2})$. Infinity is limit point since the potential is integrable on tails. Weyl's endpoint theorem therefore gives deficiency indices $(1,1)$. The boundary Wronskian is

$$W_0(\bar{f}, v) = 2\mathrm{i}\sigma(\overline{b_f} b_v - \overline{a_f} a_v).$$

Its maximal isotropic lines are exactly $a = \kappa b$, $|\kappa| = 1$; neither coefficient of a nonzero isotropic vector can vanish. Conjugation changes the ratio to $1/\overline{\kappa} = \kappa$. For nonsemiboundedness, set $f(x) = x^{1/2}\chi(\log x)$ with a real compactly supported smooth function χ. Direct substitution gives

$$\langle f, \ell f\rangle = \int_{\mathbb{R}} (|\chi'(s)|^2 - \sigma^2|\chi(s)|^2)\,\mathrm{d}s.$$

A sufficiently long plateau makes this negative. Normalized dilations make it arbitrarily negative. Every extension contains these minimal-domain test functions, so no semibounded extension exists. □

Put $\varsigma = \kappa\Gamma(-\mathrm{i}\sigma)/\Gamma(\mathrm{i}\sigma) = \mathrm{e}^{\mathrm{i}\theta}$, using any real lift θ. Powers of a right-half-plane momentum use the logarithm on $\operatorname{Re}\rho > 0$.

Theorem 2 (The entire negative spectrum)**.** *Every negative spectral point of* $H_{\sigma,\kappa} = \ell|_{D_\kappa}$ *is a simple eigenvalue*

$$E_j = -\rho_j^2 = -4\exp\left(-\frac{\theta + 2\pi j}{\sigma}\right), \quad j \in \mathbb{Z}, \qquad \psi_j(x) = \rho_j\sqrt{\frac{2\sinh(\pi\sigma)}{\pi\sigma}}\sqrt{x}K_{\mathrm{i}\sigma}(\rho_j x). \tag{3}$$

The functions ψ_j *have unit norm. A different lift of* θ *merely reindexes the same spectrum.*

Proof. At energy $-\rho^2$ the decaying solution at infinity is uniquely $\sqrt{x}K_{\mathrm{i}\sigma}(\rho x)$, up to a scalar. Its endpoint coefficients follow from

$$K_{\mathrm{i}\sigma}(w) = \tfrac{1}{2}\Gamma(\mathrm{i}\sigma)(w/2)^{-\mathrm{i}\sigma} + \tfrac{1}{2}\Gamma(-\mathrm{i}\sigma)(w/2)^{\mathrm{i}\sigma} + o(1).$$

Their ratio equals κ exactly when $\varsigma(\rho/2)^{2\mathrm{i}\sigma} = 1$, giving (3). For $\operatorname{Re}\rho > 0$ this equality forces $\arg\rho = 0$ by taking absolute values. Away from these roots the Green kernel in the proof supplement gives a bounded inverse; hence no additional negative spectrum is omitted. Uniqueness of the decaying solution gives simplicity.

For $f_a(x) = \sqrt{x}K_{\mathrm{i}\sigma}(ax)$ and $f_b(x) = \sqrt{x}K_{\mathrm{i}\sigma}(bx)$, the Lagrange identity is $W(f_a, f_b)' = (b^2 - a^2)f_a f_b$. The Wronskian vanishes at infinity; its endpoint value is $\sigma|\Gamma(\mathrm{i}\sigma)|^2\sin(\sigma\log(a/b))$. Using $|\Gamma(\mathrm{i}\sigma)|^2 = \pi/(\sigma\sinh\pi\sigma)$ gives

$$\int_0^\infty xK_{\mathrm{i}\sigma}(ax)K_{\mathrm{i}\sigma}(bx)\,\mathrm{d}x = \frac{\pi\sin(\sigma\log(a/b))}{(a^2 - b^2)\sinh(\pi\sigma)}. \tag{4}$$

The coincident limit is $\pi\sigma/(2a^2\sinh\pi\sigma)$, proving the normalizer in (3). Endpoint $O(x)$ bounds and exponential tail decay justify that limit. Distinct ladder values give orthogonality in (4). Finally $E_j \to 0$ as $j \to +\infty$ and $E_j \to -\infty$ as $j \to -\infty$; there is no lowest level. □

3 Green jumps, completeness and scattering

For $z = -\rho^2$, $\operatorname{Re}\rho > 0$, define

$$T_\rho = \varsigma(\rho/2)^{2\mathrm{i}\sigma},$$
$$u_\rho(x) = \sqrt{x}\,[I_{\mathrm{i}\sigma}(\rho x) - T_\rho I_{-\mathrm{i}\sigma}(\rho x)], \qquad v_\rho(x) = \sqrt{x}K_{\mathrm{i}\sigma}(\rho x). \tag{5}$$

The Gamma recurrence gives the required endpoint ratio for u_ρ. Both I–K Wronskians equal -1, so $W(u_\rho, v_\rho) = T_\rho - 1$. Thus

$$G_\kappa(-\rho^2; x, y) = \frac{u_\rho(\min(x,y))v_\rho(\max(x,y))}{1 - T_\rho}. \tag{6}$$

Its first derivative jumps by -1. The $O(\sqrt{xy})$ behavior on compact endpoint squares and exponential tail bounds give a bounded inverse when the denominator is nonzero.

For positive k, put $t(k) = \varsigma(k/2)^{2\mathrm{i}\sigma}$, $a = \mathrm{e}^{\pi\sigma/2}$ and $b = \mathrm{e}^{-\pi\sigma/2}$. Define

$$\phi_k(x) = \mathrm{e}^{-\mathrm{i}\pi/4}\sqrt{kx}\,\frac{J_{\mathrm{i}\sigma}(kx) - t(k)J_{-\mathrm{i}\sigma}(kx)}{b - t(k)a}. \tag{7}$$

The denominator satisfies $|b - ta| \geq a - b > 0$, uniformly in k at fixed σ. The two Green boundary values give

$$\frac{k}{\pi\mathrm{i}}[G_\kappa(k^2 + \mathrm{i}0; x, y) - G_\kappa(k^2 - \mathrm{i}0; x, y)] = \phi_k(x)\overline{\phi_k(y)}. \tag{8}$$

Theorem 3 (Complete spectral measure)**.** *The positive spectrum is absolutely continuous of multiplicity one, with momentum density $\phi_k(x)\overline{\phi_k(y)}\,\mathrm{d}k$. There are no positive or zero eigenvalues and no singular continuous spectrum. For every $f \in \mathcal{H}$,*

$$\|f\|^2 = \sum_{j\in\mathbb{Z}} |\langle\psi_j, f\rangle|^2 + \int_0^\infty |\langle\phi_k, f\rangle|^2\,\mathrm{d}k. \tag{9}$$

The transform is initially defined weakly on compactly supported test functions and extended by this identity.

Proof. The I–J and K–Hankel identities in (6) give (8). On compact positive momentum intervals the boundary values are continuous between the weighted spaces used in limiting absorption, with weight exponent greater than $1/2$; endpoint and oscillatory tail bounds apply because $b - ta$ never vanishes. These are the nonexceptional hypotheses in [1, Theorem 6.1 and Proposition 6.7]. Stone's formula gives the displayed density and rules out singular spectrum on every such interval. Positive-energy Bessel waves have nondecaying oscillatory tails, so no nonzero solution is square integrable there. For every nonzero zero-energy combination, dilation by $\mathrm{e}^{\pi/\sigma}$ multiplies its squared norm on consecutive logarithmic annuli by $\mathrm{e}^{2\pi/\sigma} > 1$, so it is not square integrable at infinity. A singular measure supported only at zero would be an eigenvalue atom, which is excluded. The full spectral theorem together with the negative resolvent classification now yields (9). One generalized eigenfunction per momentum gives multiplicity one. $\square$

The large-x expansion, with no unspecified normalization phase, is

$$\phi_k(x) = \frac{\mathrm{e}^{-\mathrm{i}kx} + R(k)\mathrm{e}^{\mathrm{i}kx}}{\sqrt{2\pi}} + O(x^{-1}), \qquad R(k) = -\mathrm{i}\frac{a - t(k)b}{b - t(k)a}. \tag{10}$$

Incoming means $\mathrm{e}^{-\mathrm{i}kx}$. Numerator and denominator in the ratio have equal modulus, so $|R| = 1$. This is reflection, not yet relative scattering. The free Dirichlet half-line Hamiltonian H_D has reflection -1. Under the convention

$$W_\pm = \operatorname*{s-lim}_{t\to\pm\infty} \mathrm{e}^{\mathrm{i}tH_{\sigma,\kappa}}\mathrm{e}^{-\mathrm{i}tH_D}, \qquad S = W_+^* W_-,$$

the sine-representation multiplier is $S(k) = -R(k)$. Existence and completeness of these limits are the explicitly used external result [1, Proposition 6.10]. Its two parameter pairs are $(\mathrm{i}\sigma, \kappa)$ and $(1/2, 0)$: both real parts lie strictly between minus one and one, both realizations are self-adjoint, and neither denominator is exceptional. Proposition 6.9 of that source agrees with the phase in (10). Completeness means $W_\pm^* W_\pm = I$ and $W_\pm W_\pm^* = P_{\mathrm{ac}}(H)$, not that negative bound states scatter freely.

4 Reproducibility and literature ownership

The exact producer contains 45 classical phase points, 27 boundary-flux rows and 15 algebraic scattering rows. A non-importing SymPy checker reconstructs their values and literal scalar types. Sixty bound-level and 36 continuum rows are stored to 60 digits after 100-digit evaluation and independently recomputed at 110 digits through endpoint-normalized coefficients. The separate symbolic lane checks eight identities. The 100-digit special-function lane checks 108 actual Stone-jump cells, 12 Green Wronskians, 12 differential-equation residuals, 12 endpoint matches, three bound-state normalization integrals and 36 logarithmic-period cells. In particular, testing $|R| = 1$ alone is not treated as a spectral-density test. These finite grids are regression, not interval certification or a proof of spectral completeness. Two unrelated working-directory replays, repaired-hash attacks, strict YAML gates and smoke tests are distributed with the proof. The release rebuilds all three drafts in fresh directories with frozen epoch 1788566400 and retains actual settled compiler logs.

The external mathematical ownership is concentrated in one directly inspected primary source [1]; listing its journal and preprint as two independent works would be misleading. The paper does not claim priority for the self-adjoint family, geometric ladder or scattering formulas. Its package-level advance is the joined, auditable boundary between that complete source theorem and a target spectral construction it does not provide. The neighboring repository models are a Friedrichs Aharonov–Bohm Hamiltonian, a repulsive many-body Calogero–Moser system and a positive isotonic oscillator. They do not contain this strictly supercritical, imaginary-order, all-domain bilateral ladder. The proof and source audit preserve those owner differences.

Local collaborating AI agents reviewed the proof and its normalization; this is not external human peer review or cross-provider validation. An independent reading prompted direct Green/Stone tests and a fully expanded normalization identity, both implemented here. No outside model was invoked.

5 Conclusion

Round one: complete continuum and scattering. The Green boundary jump and endpoint analysis close both spectral components. The normalized reflection and relative scattering phases remain distinct, and the precise external wave-operator theorem supplies time-dependent completeness only on the absolutely continuous subspace.

References

[1] J. Dereziński and S. Richard. On Schrödinger operators with inverse square potentials on the half-line. *Annales Henri Poincaré* **18**, 869–928 (2017). doi:10.1007/s00023-016-0520-7. Author manuscript: arXiv:1604.03340v2. The theorem numbering used here follows that inspected preprint.